WAHA vs Evolution API

Comparison & recommendation (WhatsApp POC)

Overview

Both POCs share the same Flask product flow: login → QR connect → sync contacts → compose/campaign. Only the WhatsApp engine behind the API differs.

Shared Flask flow

Step	Flow
1	Login admin / admin123
2	Connect — start session/instance, scan QR
3	Contacts — sync from WhatsApp
4	Compose — free-text message
5	Campaign — queue send to selected contacts

POC folders & ports

Folder	Engine	Flask app	Engine port
poc/waha-poc/	WAHA	http://127.0.0.1:5001	3000
poc/evolution-poc/	Evolution API	http://127.0.0.1:5002	8080

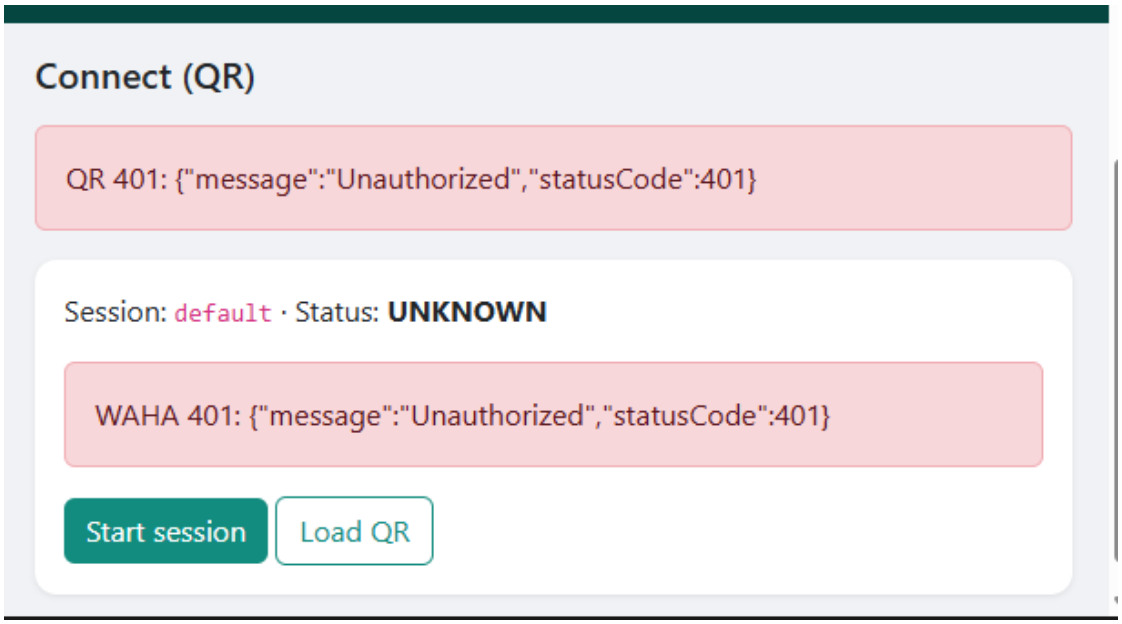
Admin UIs (screenshots)

WAHA — built-in dashboard

URL: http://127.0.0.1:3000/dashboard/

Auth: Username + password from waha-env/.env; link worker with WAHA_API_KEY.

WAHA dashboard — workers and sessions

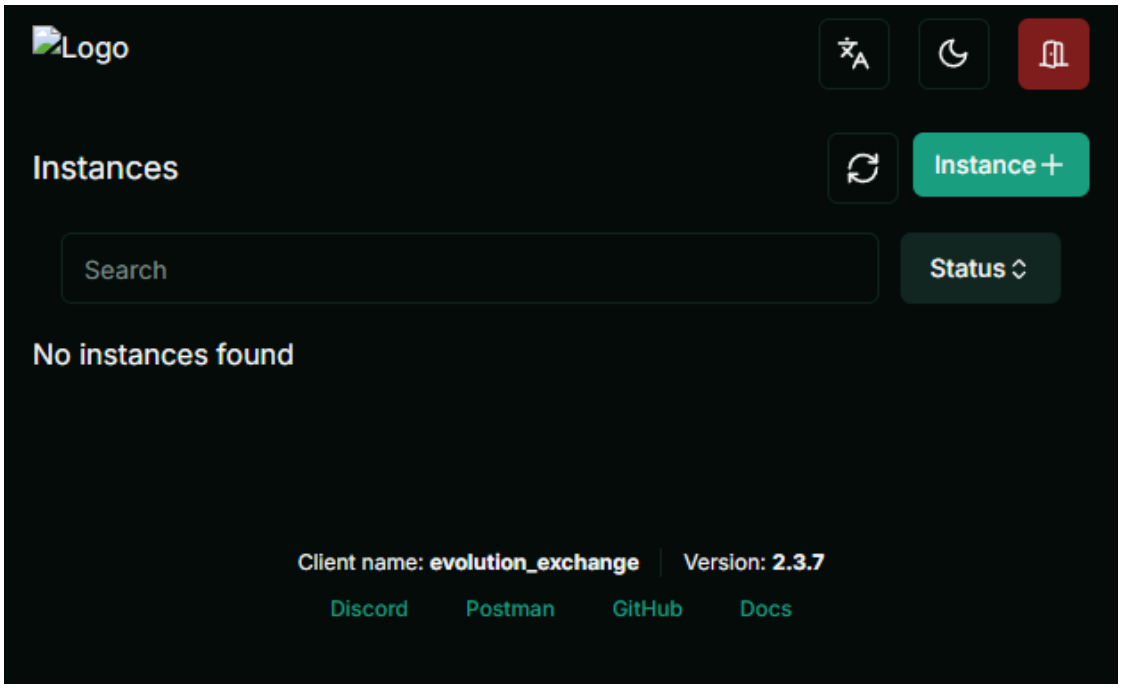


Evolution — Evolution Manager

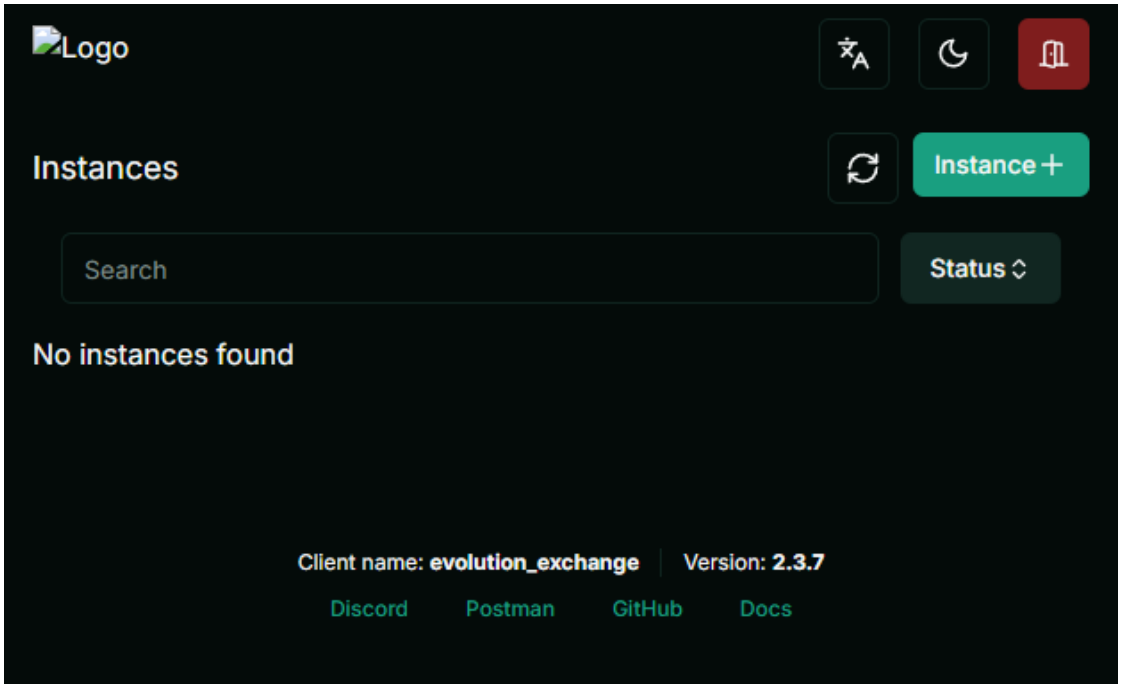
URL: http://127.0.0.1:8080/manager

Auth: Global API key only (same as EVOLUTION_API_KEY).

Evolution Manager login



Evolution Manager — instances



Flask demo (both POCs)

Same UX on ports 5001 (WAHA) and 5002 (Evolution): Connect, Contacts, Compose, Campaigns. Use for stakeholder demos; use native UIs for engine debugging.

Side-by-side comparison

Criteria	WAHA	Evolution API
Docker footprint	1 container (WAHA only)	3 containers (API + Postgres + Redis)
First-time setup	init-waha.bat + sync API key to Flask .env	Matching API key in compose + .env; longer first pull
Admin login	Username + password + API key for worker	API key only on Manager
Session model	Named sessions (default)	Named instances (default)
QR connect	Works; 422 if QR while STARTING	Works; create/start instance first
Contact sync	Straightforward API in our client	v2 POST findContacts with {"where":{}}
API auth header	X-Api-Key	apikey
Multi-instance / SaaS	Sessions + namespace	Strong fit — DB, Manager, Redis
Best for one number bulk	Excellent	Good

Best for many numbers	Good	Better

POC scorecard

Criteria	WAHA	Evolution
Install time	~5 min (after Docker)	~10–15 min (DB + Redis + image)
QR connected	Yes	Yes
Contact sync	Yes	Yes (after POST fix)
Single send	Yes	Yes
Campaign queue	Yes	Yes
Stability (short demo)	Stable	Stable
Docs / ease for devs	Easier (fewer services)	More moving parts
Native admin UX	Workers/sessions dashboard	Polished Manager
Pick for this product	Recommended	Strong alternate

Legal & operations (both)

- Uses WhatsApp Web / Baileys-class linking — not the official Meta Cloud API.
- Free-text is possible but policy and ban risk still apply.
- Use a dedicated test number, rate limits, and opt-in/consent for marketing.
- For compliance-first production, evaluate legacy-cloud-api/ (Cloud API) separately.

Recommendation: what should we use?

Primary choice: WAHA

For one web app, one WhatsApp number, QR login, contact sync, and bulk/campaign sends, WAHA is the better default engine:

1. Simpler operations — one Docker service, no Postgres/Redis on a laptop or small VPS.
2. Faster onboarding — init-waha.bat + one API key; clear 401 when keys mismatch.
3. Enough admin UI — dashboard + Swagger for support staff.
4. Lower POC friction after credentials were fixed.

When to choose Evolution instead

- Many WhatsApp instances (multi-tenant) with Manager as ops console.
- Team already on Evolution webhooks or evoapicloud in production.
- Need Evolution ecosystem (chatbots, n8n, etc.) without rebuilding on WAHA.

Decision summary

Question	Answer
Which engine for v1 product?	WAHA
Which Flask POC to demo first?	http://127.0.0.1:5001
Keep Evolution POC?	Yes — reference & multi-instance evaluation
Official Meta path?	Separate decision; see legacy-cloud-api/

Next steps (optional)

5. Merge one Flask app + configurable engine (WAHA / EVOLUTION / MOCK).
6. Improve Connect page: poll until QR-ready (fix WAHA 422 UX).
7. Stakeholder sign-off using this doc + live demo on 5001.
8. Production hardening: rotate API keys, HTTPS reverse proxy, secrets not in git.

Generated from poc/COMPARISON.md — WhatsApp POC repository.